EBN 122

Chapter 5:
Operational Amplifiers (Op Amps)

Cascaded Op-Amp Circuits

Chapter 5 overview

- 5.2. Operational Amplifiers (Op Amps)
- 5.3. Ideal Op Amps
- 5.4. Inverting Amplifiers
- 5.5. Noninverting Amplifiers
- 5.6. Summing Amplifiers
- 5.7. Difference Amplifiers
- 5.8. Cascaded Amplifier Circuits

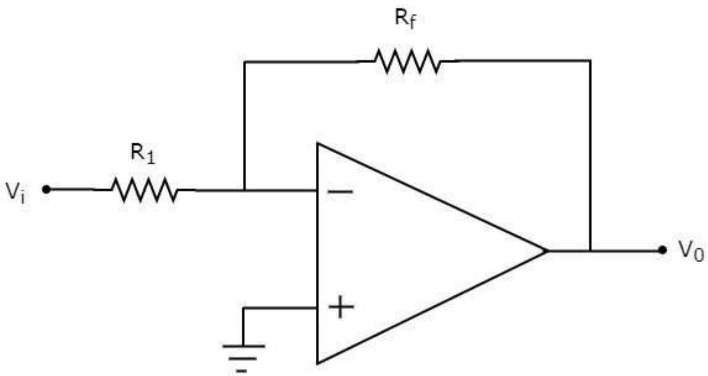
Recap of op-amps

IDEAL OP-AMPS:

$i_+ = i_- = 0$

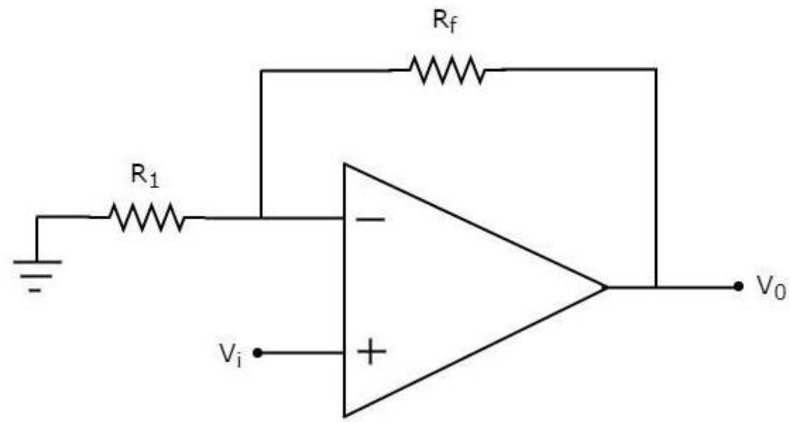
$v_+ = v_-$

Inverting:



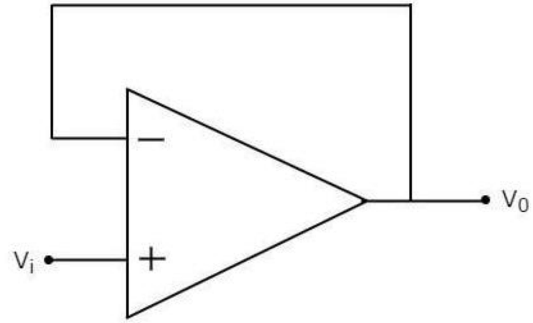
$$v_o = -\frac{R_f}{R_1} v_i$$

Non-inverting:



$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$

Buffer (voltage follower):



$$v_o = v_i$$

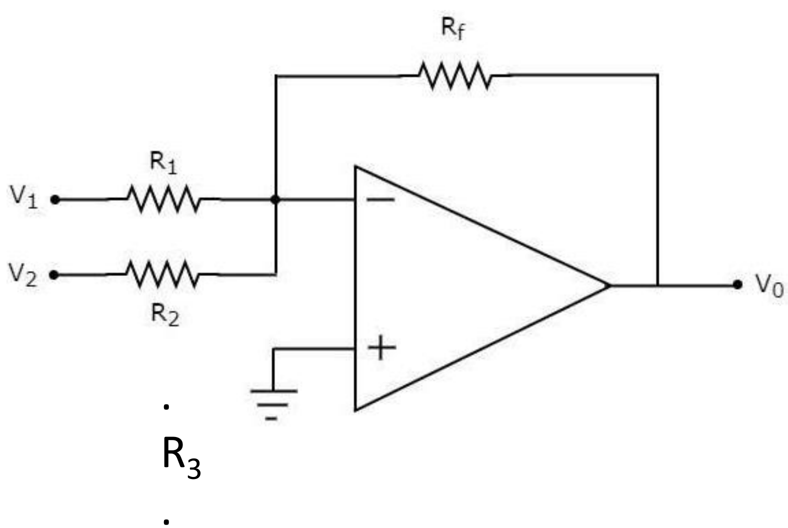
Re-cap of op-amps

IDEAL OP-AMPS:

$i_+ = i_- = 0$

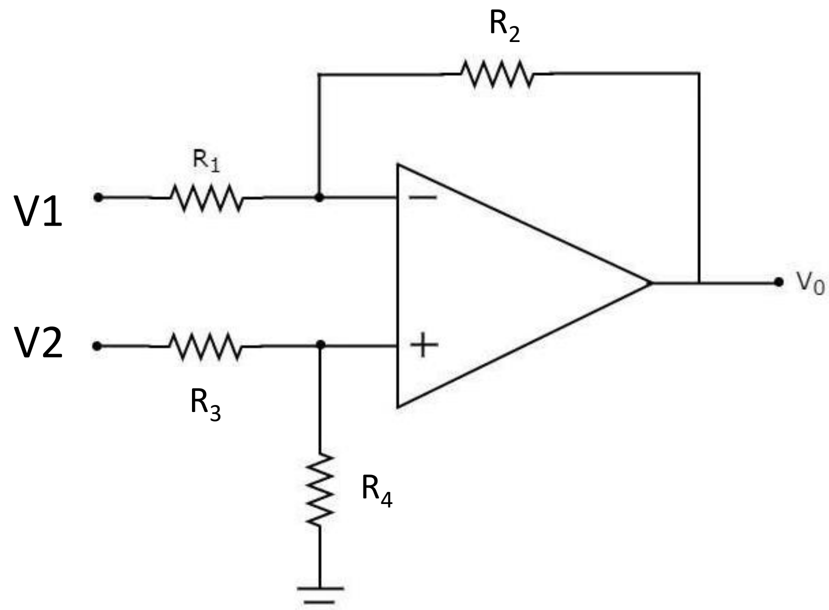
$v_+ = v_-$

Summing amplifier:



$$v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 \right)$$

Difference amplifier:



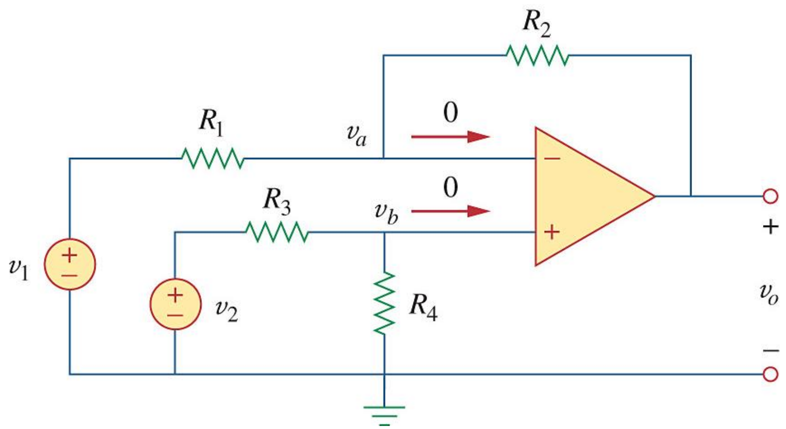
$$v_o = \frac{R_2(1 + R_1 / R_2)}{R_1(1 + R_3 / R_4)} v_2 - \frac{R_2}{R_1} v_1$$

Example 1: (Practice Problem 5.7)

Design an Op Amp circuit with inputs v_1 and v_2 such that $v_o = -5v_1 + 3v_2$

$v_o = 3v_2 - 5v_1 \rightarrow$ difference amplifier

$$\boxed{v_o = \frac{R_2(1+R_1/R_2)}{R_1(1+R_3/R_4)}v_2 - \frac{R_2}{R_1}v_1}$$



$$\frac{R_2}{R_1} = 5$$

$$\frac{R_2(1+R_1/R_2)}{R_1(1+R_3/R_4)} = \frac{5\left(1+{}^{1}/_{5}\right)}{1\left(1+{}^{R_3}/_{R_4}\right)} = 3$$

$$\frac{5\left(1+{}^{1}/_{5}\right)}{3} = \left(1+{}^{R_3}/_{R_4}\right)$$

$$2 - 1 = 1 = {}^{R_3}/_{R_4}$$

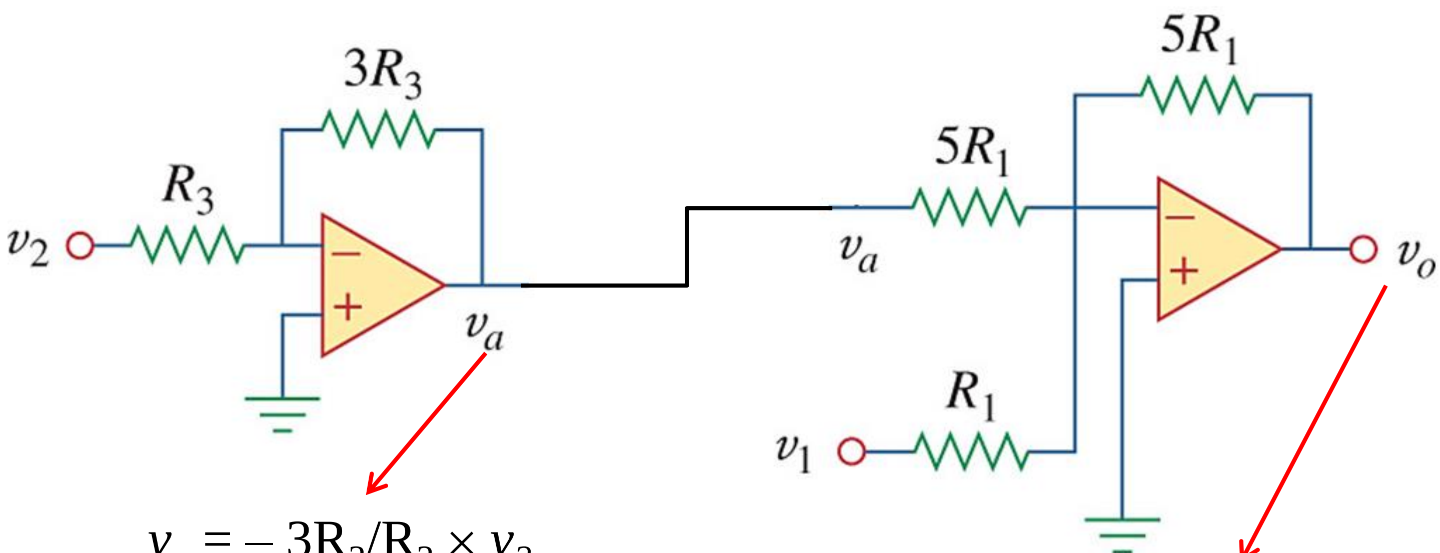
Choose $R_3 = R_4 = 2\ \text{k}\Omega$

Example 1: (Practice Problem 5.7)

Design an Op Amp circuit with inputs v_1 and v_2 such that $v_o = -5v_1 + 3v_2$

- Can also use 2 op-amps:

1. Build an inverting amp to obtain $v_a = -3v_2$.
2. Feed the output v_a into a summing amplifier along with v_1.

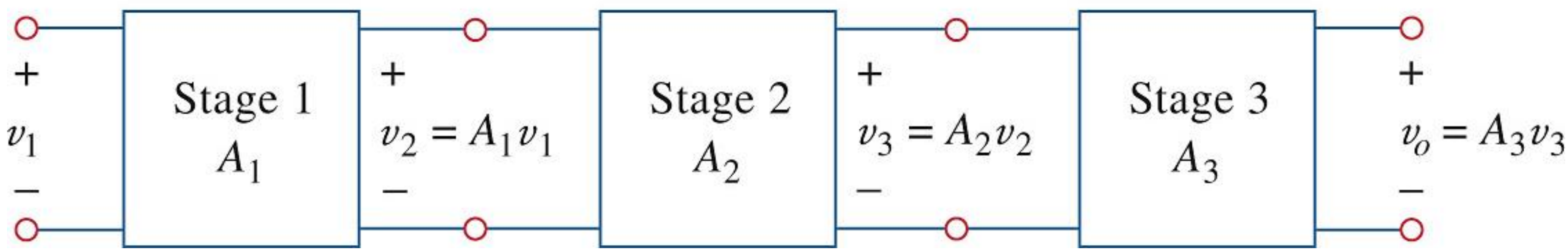


$$v_a = -3R_3/R_3 \times v_2$$
$$= -3v_2$$

$$v_o = -(5R_1/R_1 \times v_1 + 5R_1/5R_1 \times v_a)$$
$$= -5v_1 - v_a$$
$$= -5v_1 - (-3v_2)$$
$$= -5v_1 + 3v_2$$

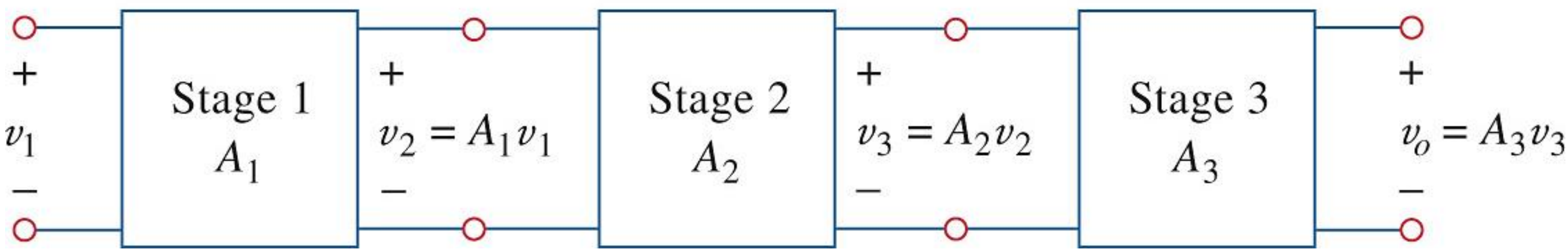
Cascaded Op Amps

- It is common to use multiple op-amp stages chained together – e.g. can increase overall gain
- This is called **cascading**
- Each amplifier is then called a **stage**



Cascaded Op Amps

- Cascaded op-amps: Head-to-tail arrangement of two or more op amp circuits such that the output to one is the input of the next

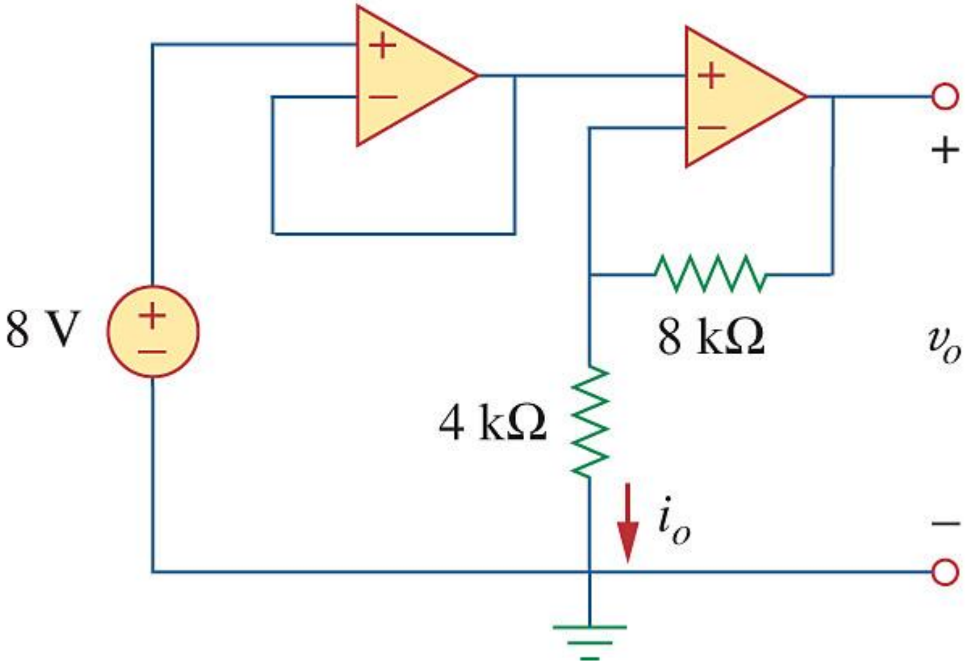


$$v_o = A_3 v_3 = A_3(A_2 v_2) = A_3 A_2 (A_1 v_1)$$

Total Amplification $A = A_1 A_2 A_3$

Example 2:

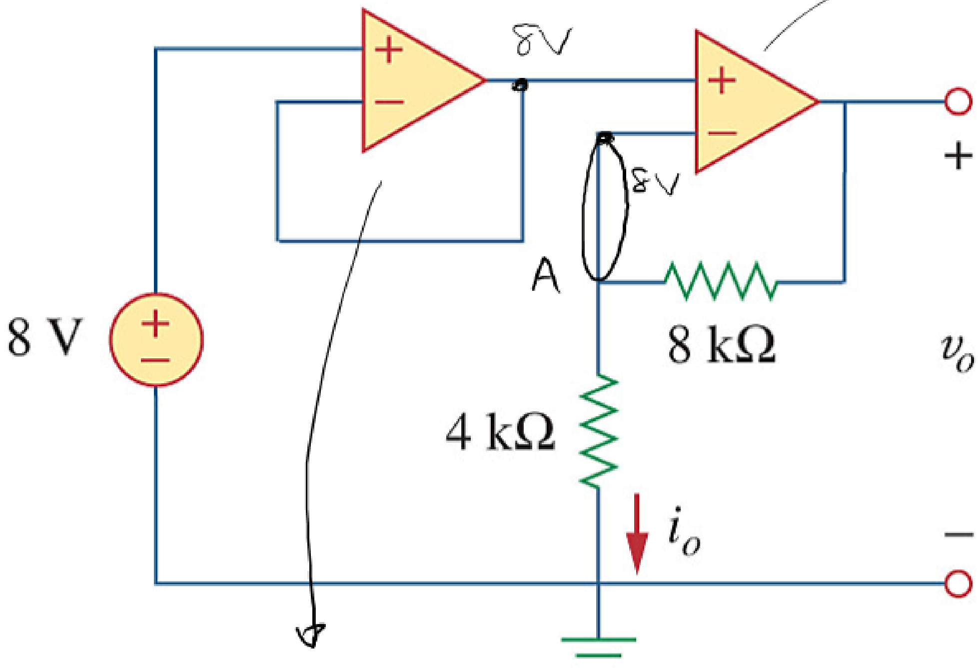
Determine v_o and i_o



Example 2:

Determine v_o and i_o

non-inverting amp



buffer

KCL at A:

$$\frac{8-0}{4k\Omega} + \frac{8-V_o}{8k\Omega} = 0$$

$$\therefore V_o = 16 + 8$$

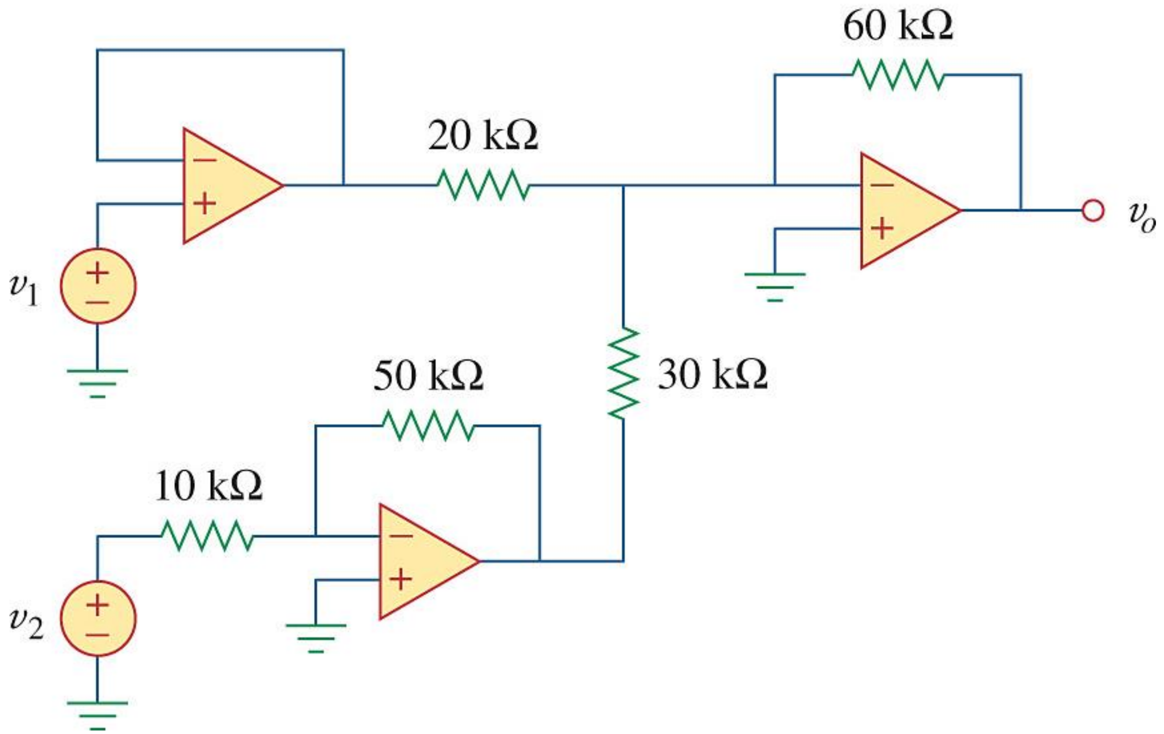
$$V_o = 24V$$

$$i_o = \frac{V_-}{4k\Omega} = \frac{8V}{4k\Omega}$$

$$i_o = 2mA$$

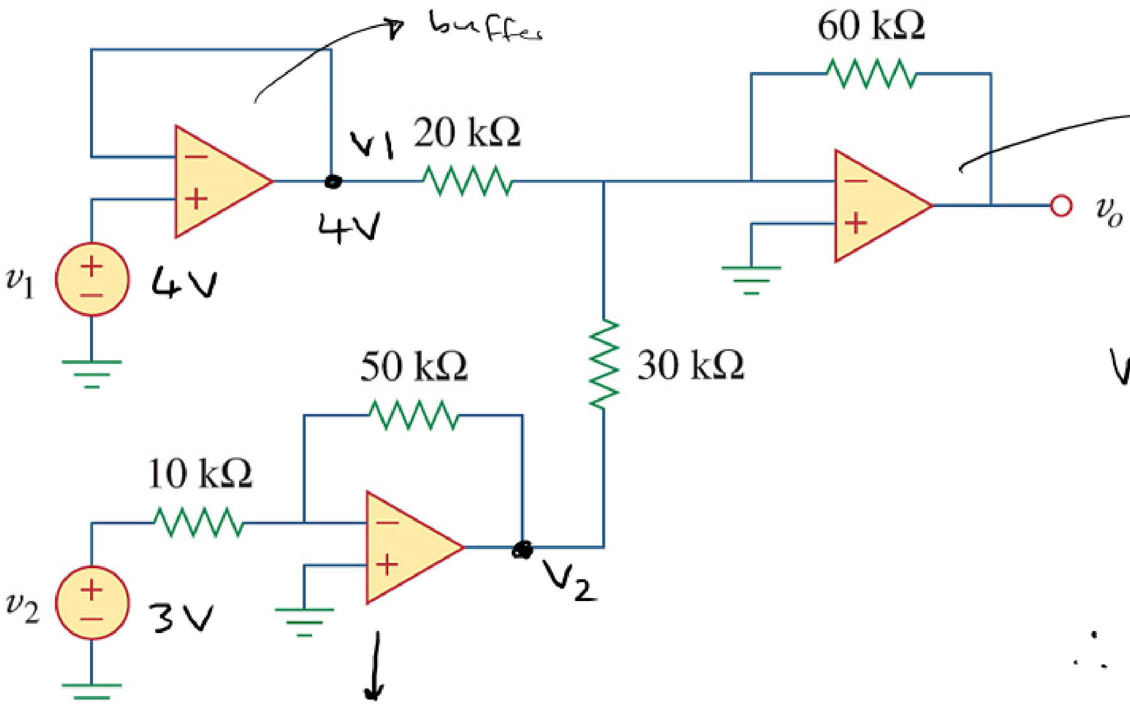
Example 3: (Practice Problem 5.10)

If v_1 = 4 V and v_2 = 3 V, determine v_o



Example 3: (Practice Problem 5.10)

If v_1 = 4 V and v_2 = 3 V, determine v_o



summing amp!

$$V_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2\right)$$

$$V_o = -\left(\frac{60k}{20k}(4V) + \frac{60k}{30k}(-15V)\right)$$

$$\therefore V_o = -12 + 30$$

$$V_o = 18V$$

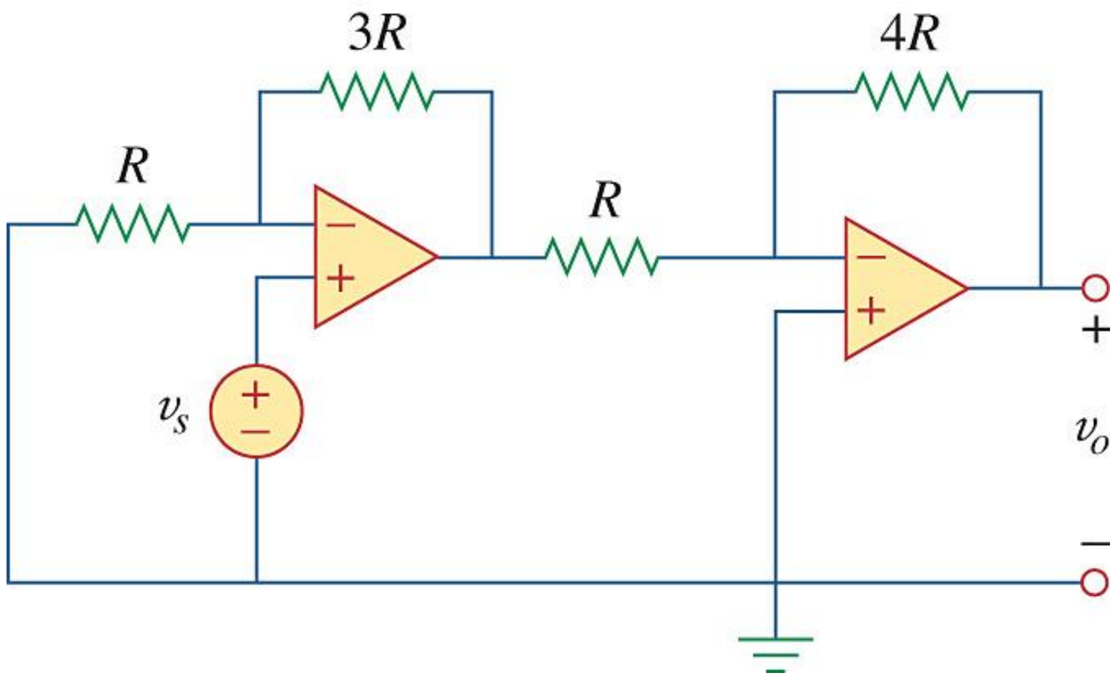
inverting

$$V_2 = -\frac{50k}{10k}(3V)$$

$$V_2 = -15V$$

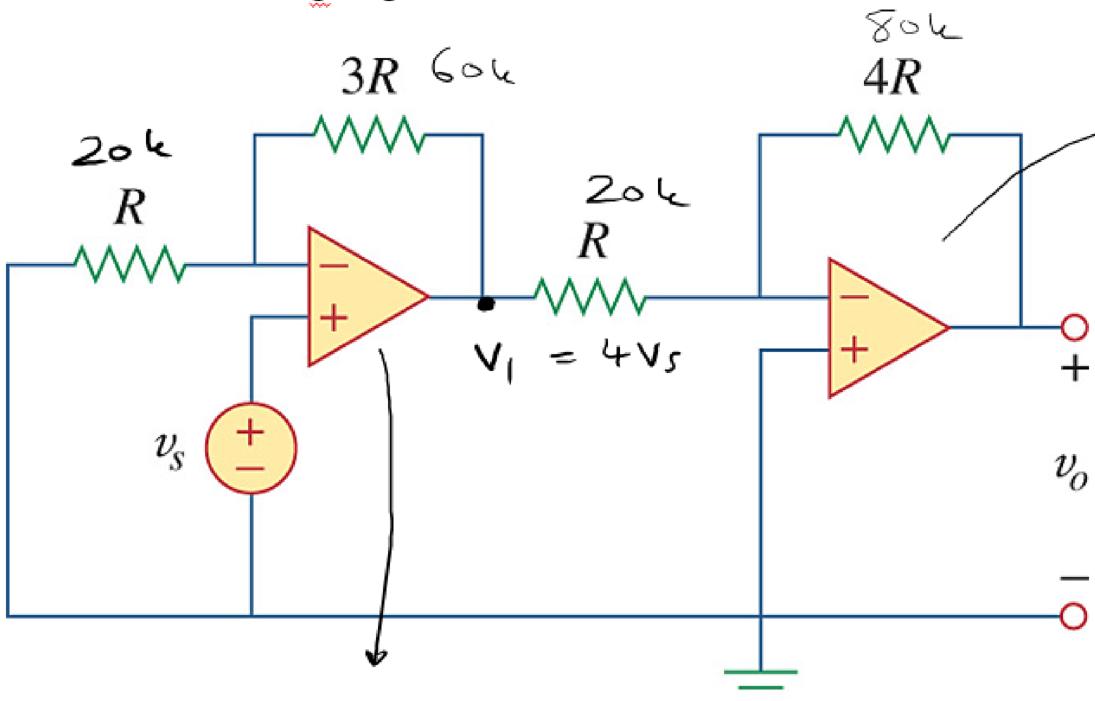
Example 4:

Determine v_o/v_s if R = 20 kΩ.



Example 4:

Determine v_o/v_s if R = 20 kΩ.



inverting

$$V_o = -\frac{R_f}{R_1} V_i$$

$$V_o = -\left(\frac{80k}{20k}\right) 4V_s$$

$$V_o = -16\ V_s$$

$$\therefore \frac{V_o}{V_s} = -16$$

non-inverting

$$V_1 = \left(1 + \frac{R_f}{R_1}\right) V_s$$

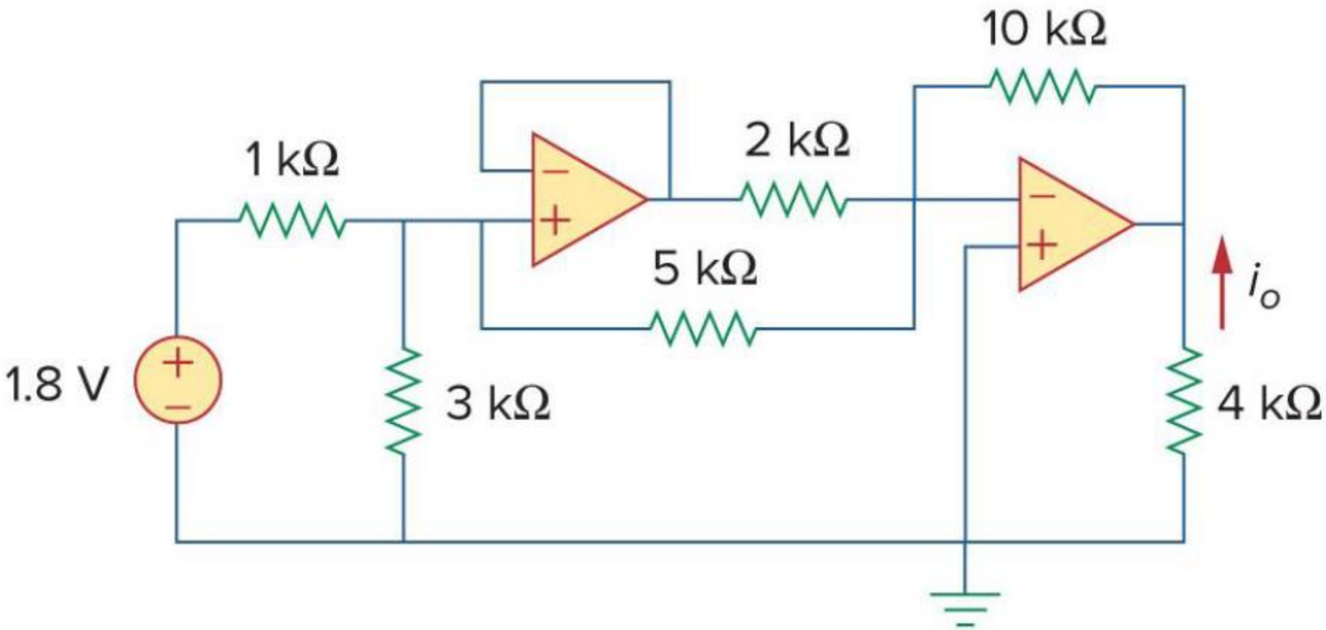
$$V_1 = \left(1 + \frac{3\cancel{R}}{\cancel{R}}\right) V_s$$

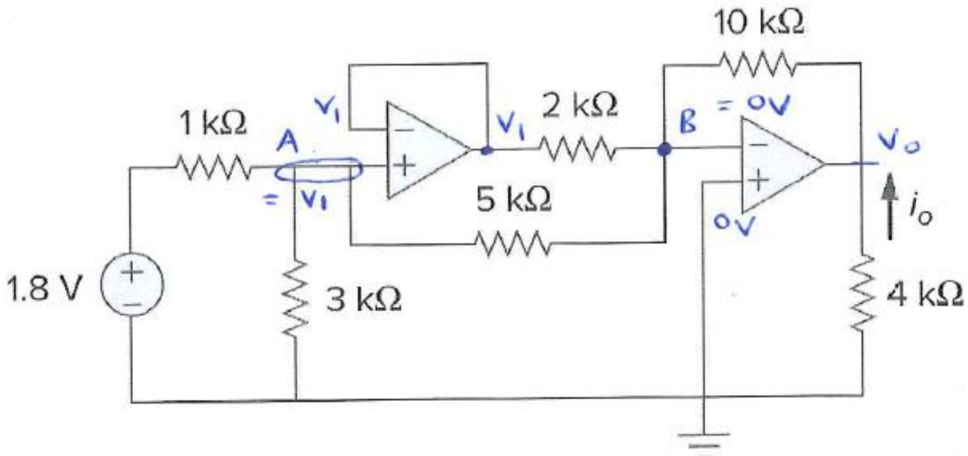
$$\therefore V_1 = 4\ V_s$$

CLASS PROBLEMS

Example 10: Cascade amplifier circuits

- Determine i_o.





With V_1 the output of the first op-amp, the input voltages $= V_1$ ∴ node $A = V_1$

With V_o the output of the second op-amp connected to 0V, node $B = 0V$

Node A: $\dfrac{V_1 - 1.8}{1k} + \dfrac{V_1}{3k} + \dfrac{V_1 - 0}{5k} = 0$

$(\times 15k)$: $15V_1 - 27 + 5V_1 + 3V_1 = 0$

$\therefore V_1 = 1.174\,V$

Node B: $\dfrac{0 - V_o}{10k} + \dfrac{0 - V_1}{2k} + \dfrac{0 - V_1}{5k} = 0$ and $V_1 = 1.174\,V$

$\therefore \dfrac{-V_o}{10} = 0.8218$

$\therefore V_o = -8.218\,V$

$i_o = \dfrac{-V_o}{4k\Omega} = \dfrac{-(-8.218)}{4k\Omega} = 0.2054\,mA$

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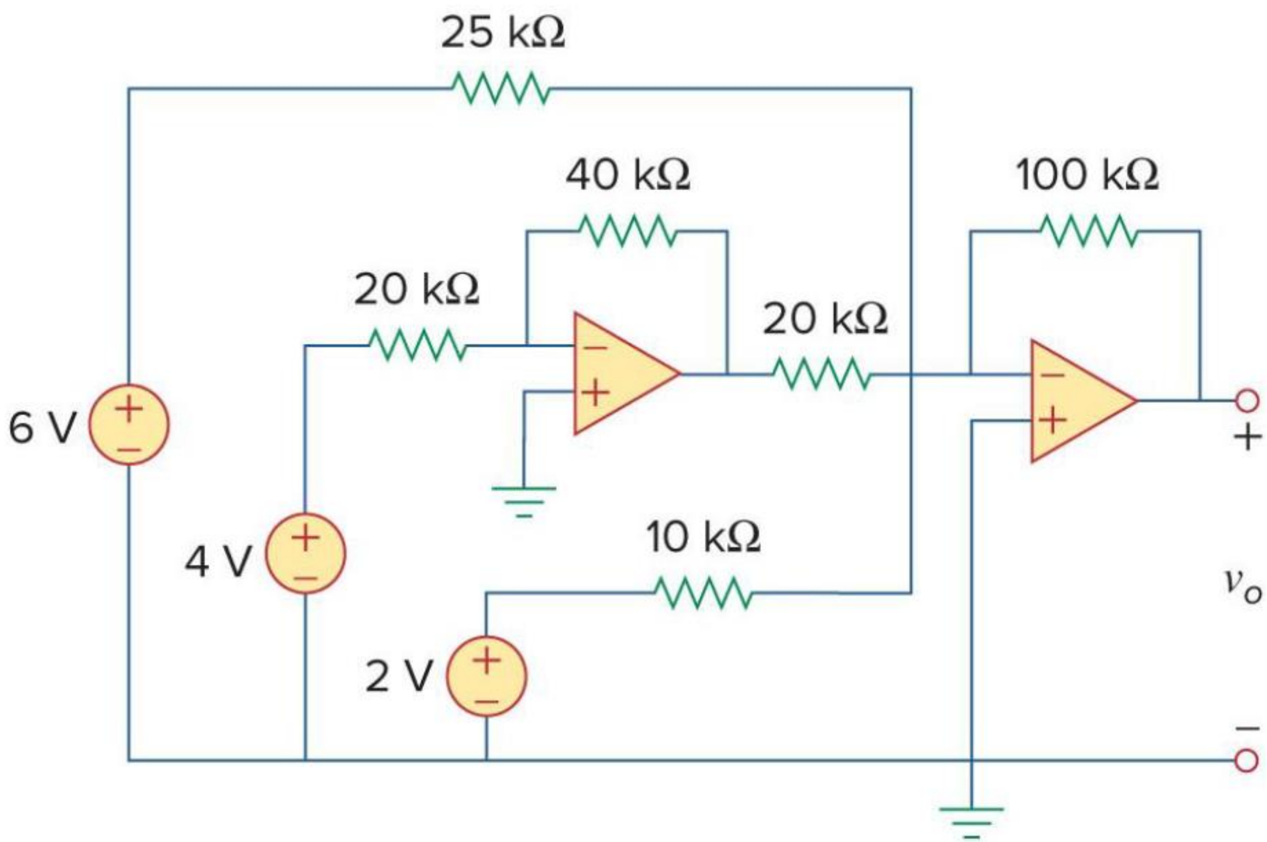
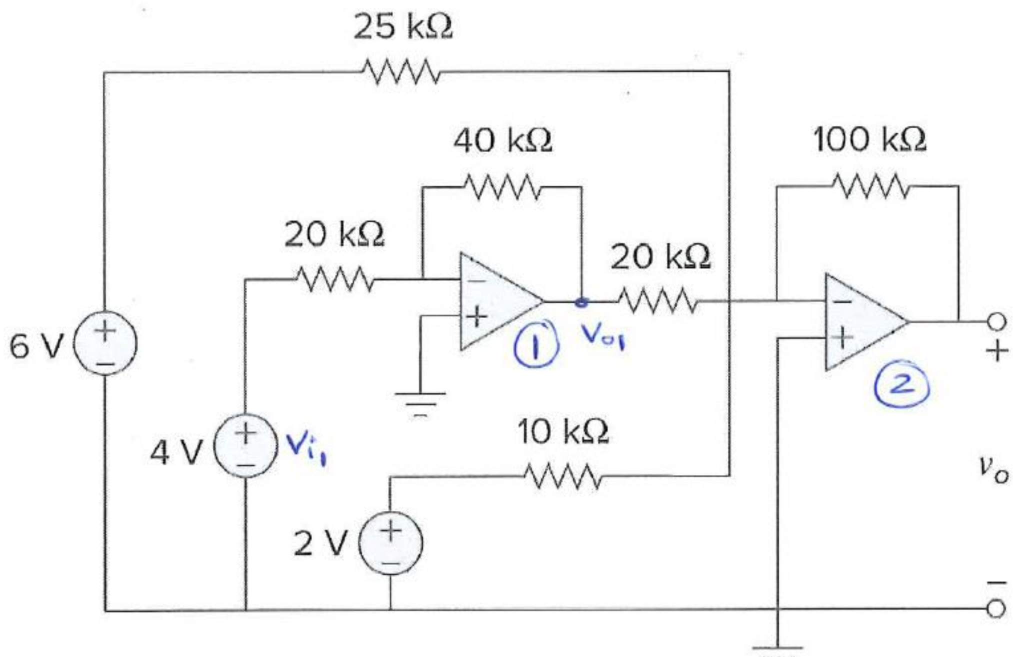
Example 11: Cascade amplifier circuits

- Determine v_o.





stage ① = inverting op-amp

stage ② = summing op-amp

① inverting:
$$V_{o1} = \frac{-R_f}{R_1} V_{i1} = \frac{-40k}{20k}(4V)$$

$$\therefore V_{o1} = -2(4) = -8V$$

② summing:
$$V_o = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)$$

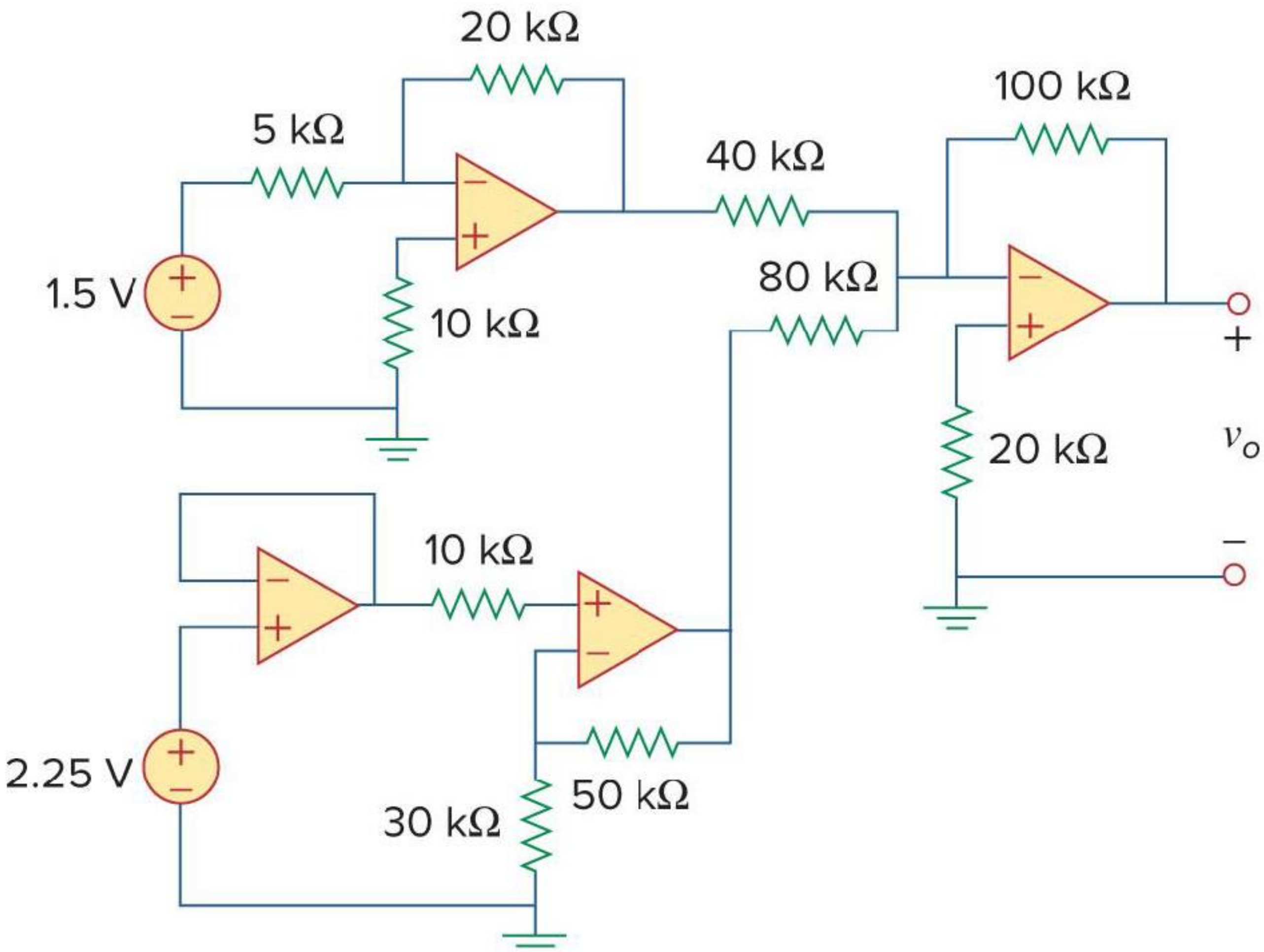
$$\therefore V_o = -\left(\frac{100k}{25k}(6) + \frac{100k}{20k}(\underset{\downarrow \atop -8}{V_{o1}}) + \frac{100k}{10k}(2)\right)$$

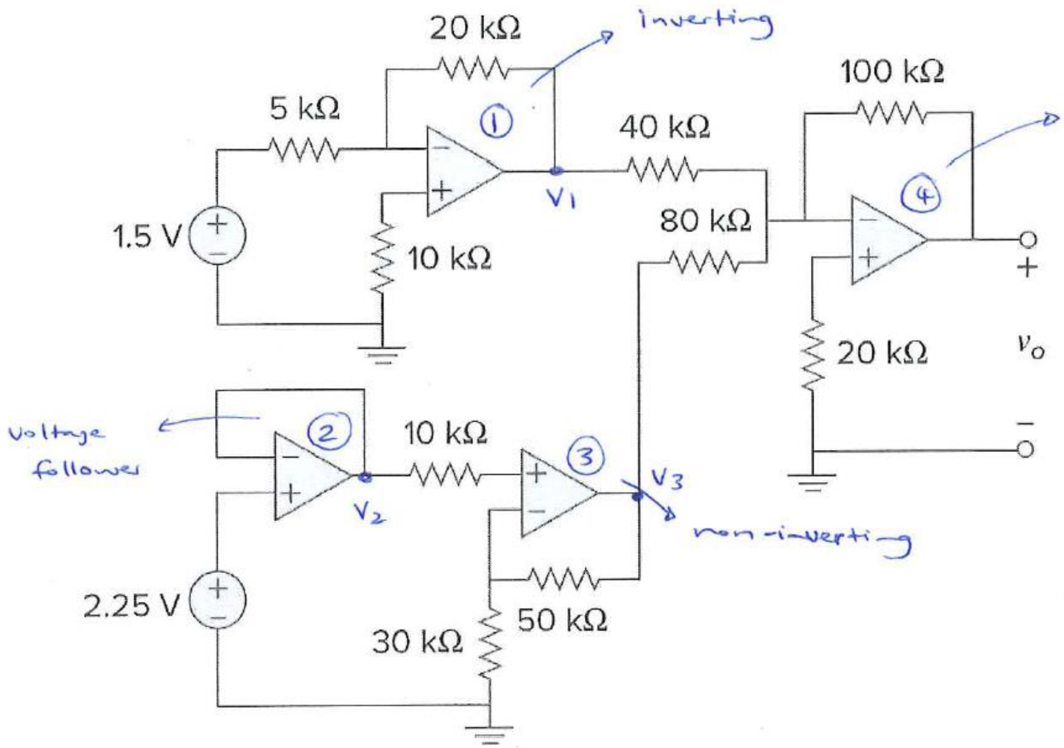
$$\therefore V_o = -(24 - 40 + 20)$$

$$\therefore V_o = -4V$$

Example 12: Cascade amplifier circuits

- Determine v_o.





$V_o \rightarrow$ summing:

$$V_o = -\left(\frac{100k}{40k} V_1 + \frac{100k}{80k} V_3\right)$$

$$V_o = -\left(2.5(-6) + 1.25(6)\right)$$

$$V_o = -\left(-15 + 7.5\right)$$

$$V_o = -(-7.5) = \underline{7.5\ V}$$

$V_1 \rightarrow$ inverting : $V_1 = \frac{-20k}{5k}(1.5V) = \underline{-6V}$

$V_2 \rightarrow$ follower : $V_2 = V_{in} = \underline{2.25\ V}$

$V_3 \rightarrow$ non-inverting: $V_3 = \left(1 + \frac{50k}{30k}\right) V_{in}$

and $V_{in} = V_2 = 2.25\ V$

$\therefore V_3 = (1 + 1.667)(2.25) = \underline{6V}$

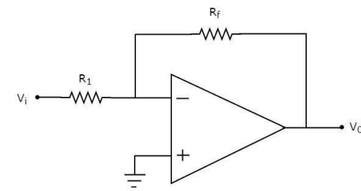
Summary

- Ideal op-amps

$i_+ = i_- = 0$

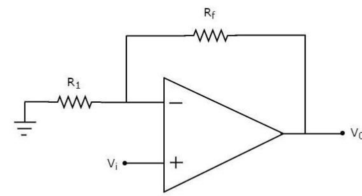
$v_+ = v_-$

- Inverting op-amps



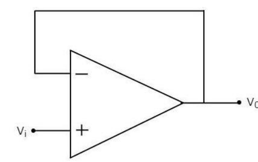
$$v_o = -\frac{R_f}{R_1} v_i$$

- Non-inverting op-amps



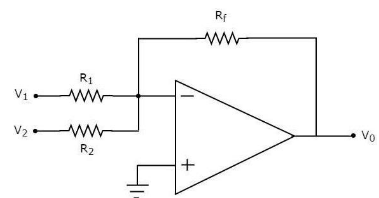
$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$

- Buffer/voltage follower



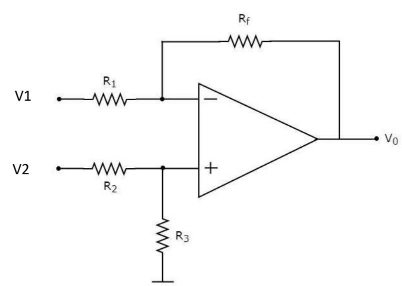
$$v_o = v_i$$

- Summing amplifier



$$v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3\right)$$

- Difference amplifier



$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

- Cascaded op-amp circuits